WEB DEVELOPMENT 1

Learning Groups: P107, P108, E101, P102

Lecturer: Mr. Maina Samuel

week 8 **learning topics**

- **Creating and Connecting Database to a PHP Registration Form**
- **8.1 Creating tables in Maria DB using SQL commands.**
- **8.2 Creating a registration form**
- **8.3 Storing data in the database**
  - **8.4 Fetching data from the Database (using CRUD operation)** `Create, Read, Update &delete`
- **Revision apart** `Create and Read` **Revise on your own** `Update &delete` **when updating a form.**

(**using SQL language**)

8.0 PHP form registration

A database is a critical tool when it comes to software development. **It helps you manage different information or records.** A database such as **MYSQL** is used by numerous developers worldwide.

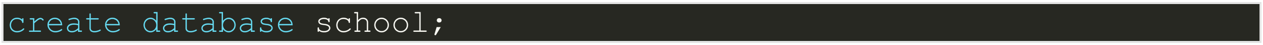
This document shows you how to use **Maria database in PHP**.

Introduction

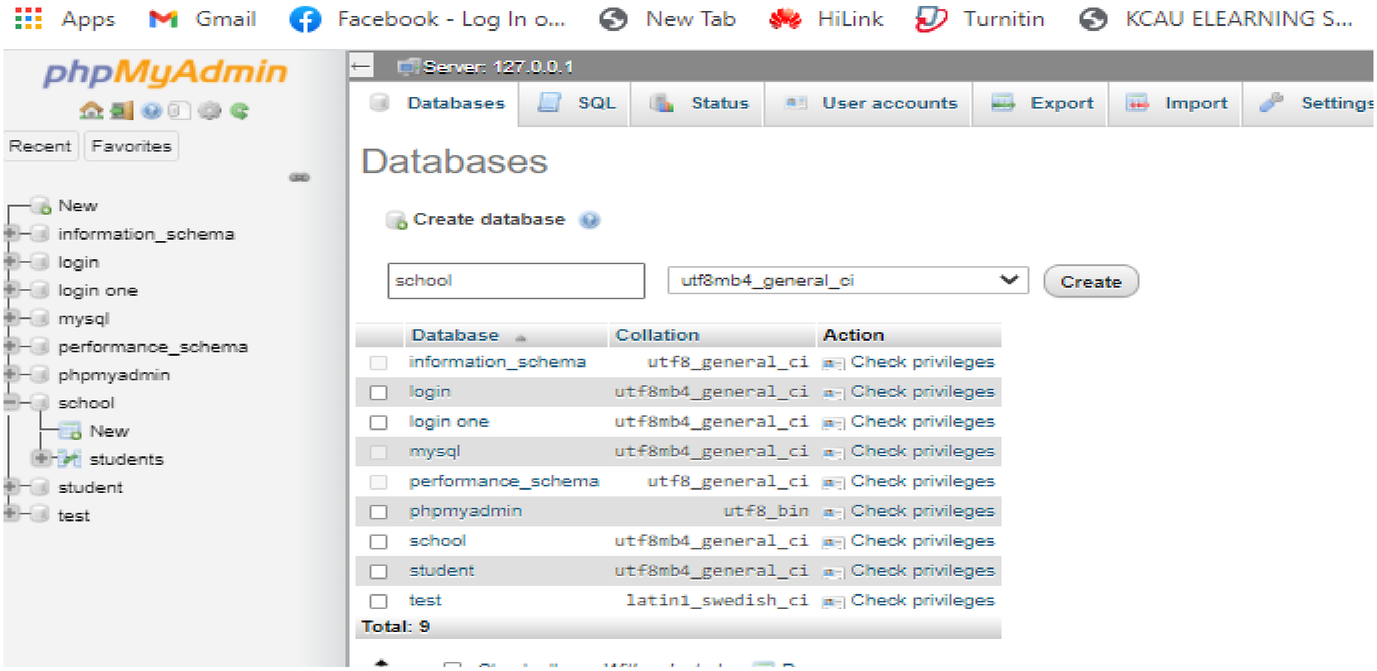
Maria Database is one of the most popular relational database management system (DBMS). It is an open source version of MySQL database. **Maria Database is based on Structured Query Language (SQL). SQL is a standard language for creating, storing, and fetching data in databases.** we are going to learn how to use Maria Database and PHP in windows.

8.1 Creating tables in Maria DB using SQL commands.

Now that we have our environment working, we can start working with Maria DB.



```
create database school;
```



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school
New
students
student
test
Server: 127.0.0.1
Databases
SQL
Status
User accounts
Export
Import
Settings
Databases
Create database
Create

school | utf8mb4_general_ci

	Database	Collation	Action
☐	information_schema	utf8_general_ci	Check privileges
☐	login	utf8mb4_general_ci	Check privileges
☐	login one	utf8mb4_general_ci	Check privileges
☐	mysql	utf8mb4_general_ci	Check privileges
☐	performance_schema	utf8_general_ci	Check privileges
☐	phpmyadmin	utf8_bin	Check privileges
☐	school	utf8mb4_general_ci	Check privileges
☐	student	utf8mb4_general_ci	Check privileges
☐	test	latin1_swedish_ci	Check privileges

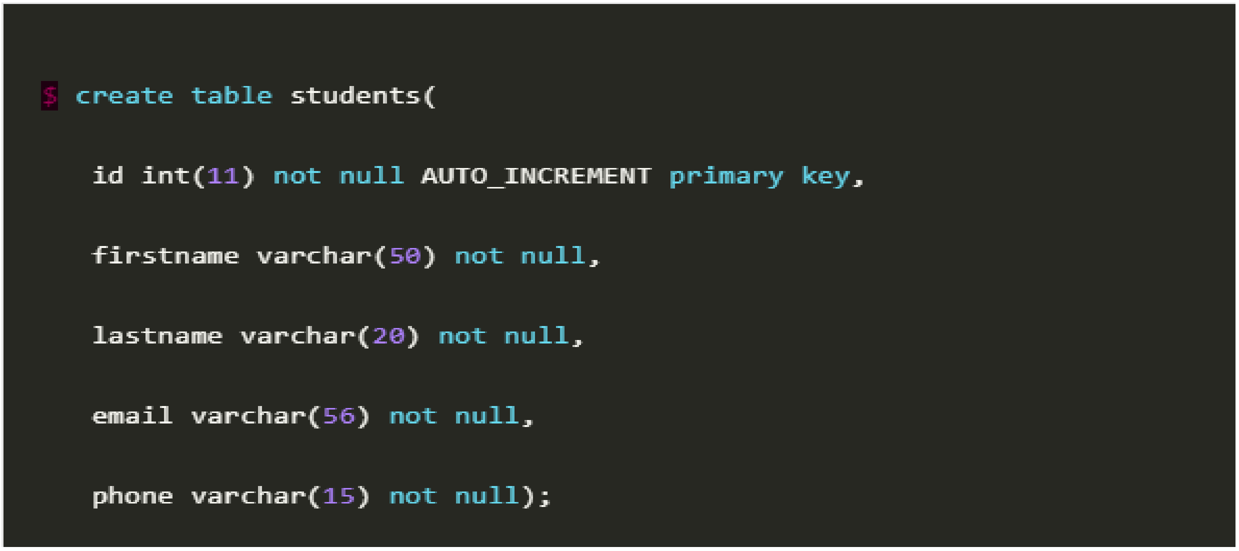
Total: 9

Step 2: select the database to use. ty command – `use school;`

`Database to be changed`

Step 3: Now, let’s create a table named students.

Use the following table commands to assist you in filling the form below it.



```
$ create table students(
    id int(11) not null AUTO_INCREMENT primary key,
    firstname varchar(50) not null,
    lastname varchar(20) not null,
    email varchar(56) not null,
    phone varchar(15) not null);
```

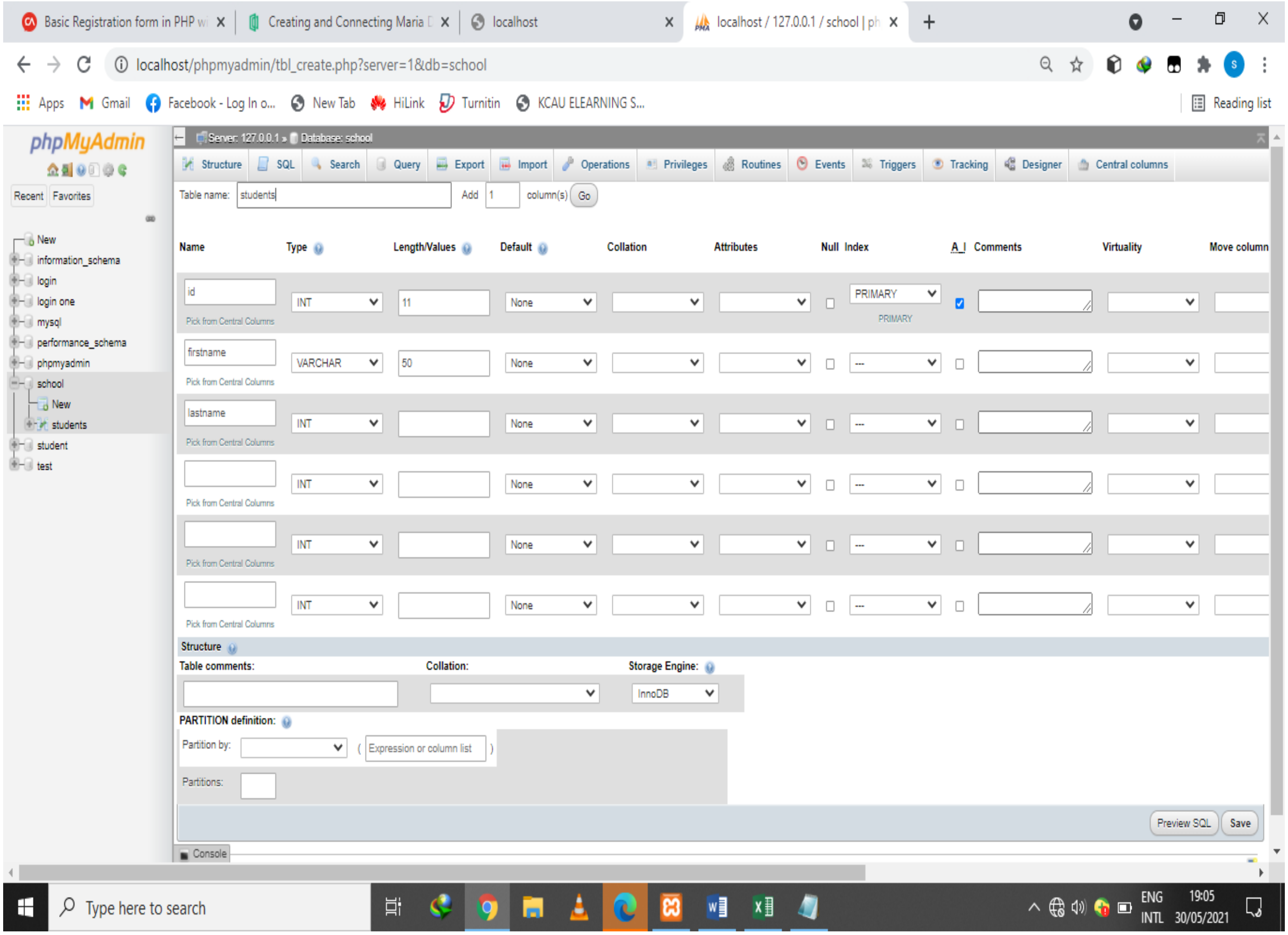

localhost/phpmyadmin/tbl_create.php?server=1&db=school
Server: 127.0.0.1 » Database: school
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
Routines
Events
Triggers
Tracking
Designer
Central columns
Preview SQL
Save

Table name: students Add 1 column(s) Go

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index	A_I	Comments	Virtuality	Move column
id	INT	11	None			☐	PRIMARY	☑			
firstname	VARCHAR	50	None			☐	---	☐			
lastname	INT		None			☐	---	☐			
	INT		None			☐	---	☐			
	INT		None			☐	---	☐			
	INT		None			☐	---	☐			

Storage Engine: InnoDB

Comments:

How to create students table: select database name on left column **>then** select **new >**table dash board will appear as illustrated on the above image

- **Fill the remaing details with the help of the table commands @ image step 3 when through click save button** located near preview SQL
- **Incase you want to add fields >click the column(s)Go button to add other fields.**

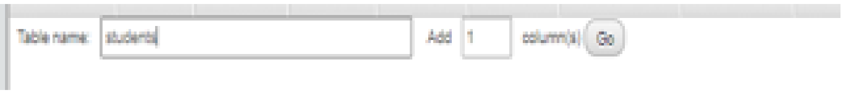
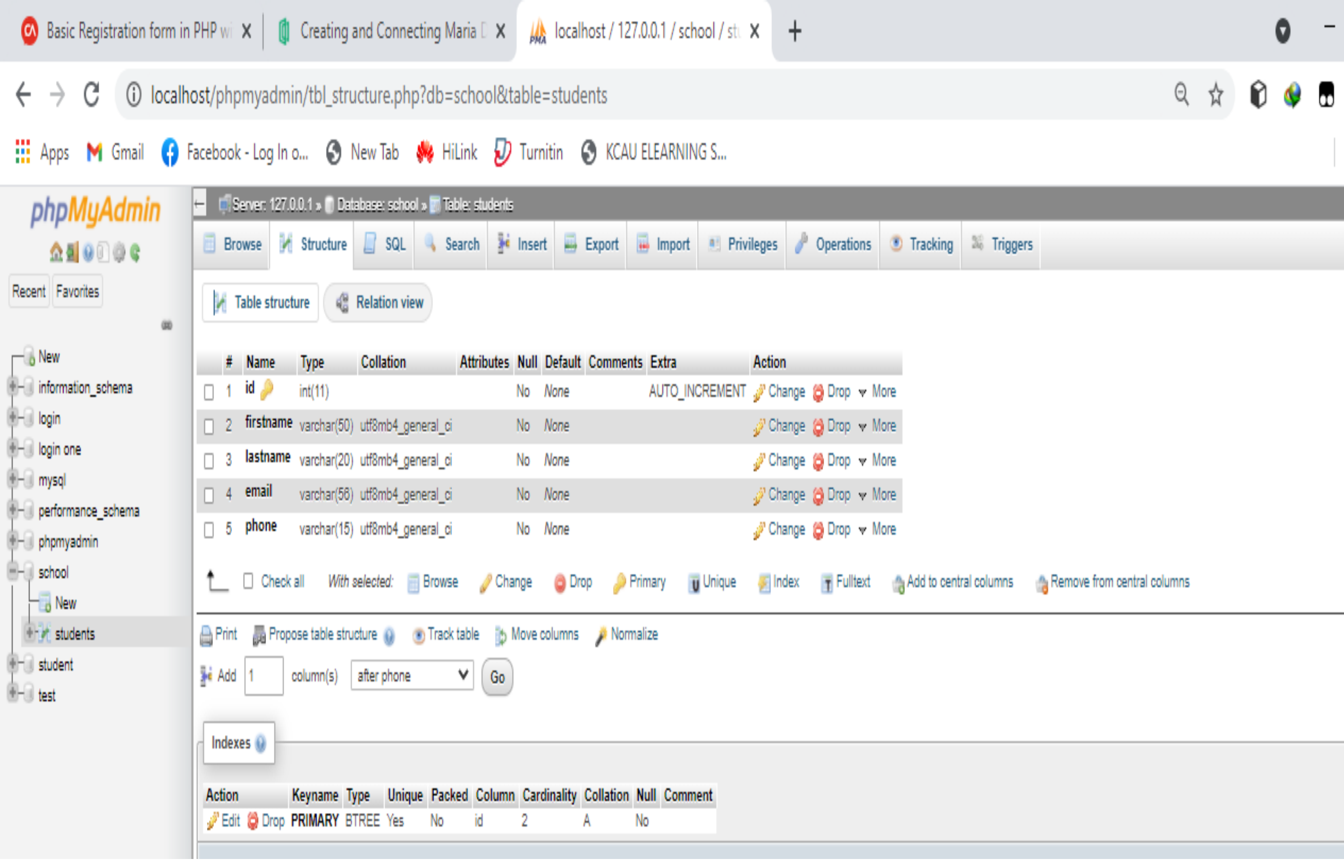


Table name: students Add 1 column(s) Go

- **To see the created table**

- So far, we have created a database school and a table student inside it. Now, let's view this database school and its tables in PhpMyAdmin. Open your browser and navigate to http://localhost/phpmyadmin/. You should see something like this.
- **Select school database Click students then structure Tab to Check table fields exist.**


Basic Registration form in PHP w
Creating and Connecting Maria D
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localhost/phpmyadmin/tbl_structure.php?db=school&table=students
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Favorites
New
information_schema
login
login one
mysql
performance_schema
phpmyadmin
school
New
students
student
test
Server: 127.0.0.1 » Database: school » Table: students
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Tracking
Triggers
Table structure
Relation view
Check all
With selected:
Browse
Change
Drop
Primary
Unique
Index
Fulltext
Add to central columns
Remove from central columns
Print
Propose table structure
Track table
Move columns
Normalize
Add
column(s)
Go
Indexes

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2	firstname	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	3	lastname	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	4	email	varchar(56)	utf8mb4_general_ci		No	None			Change Drop More
☐	5	phone	varchar(15)	utf8mb4_general_ci		No	None			Change Drop More

Add 1 column(s) after phone

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	id	2	A	No	

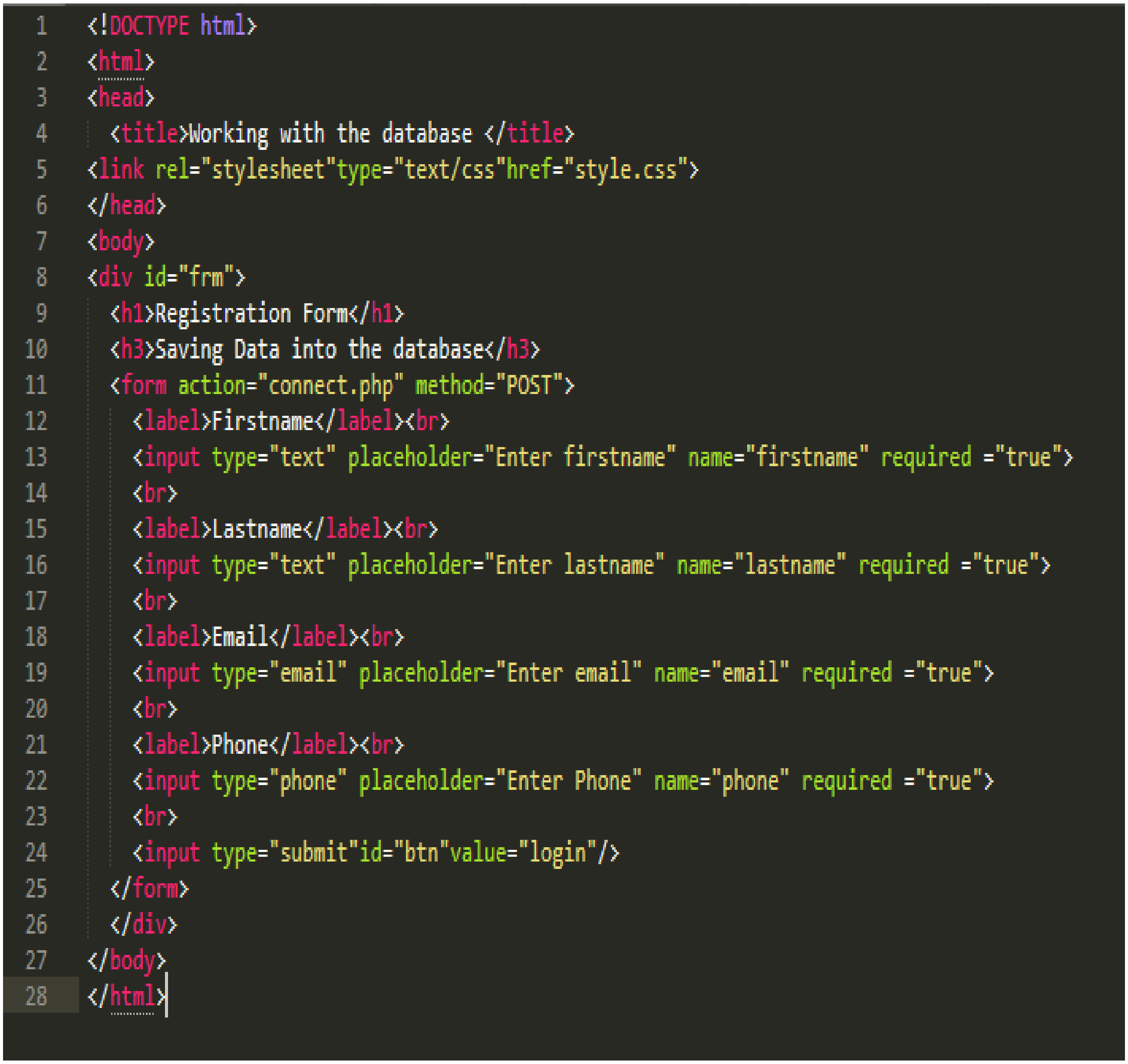
8.2 Creating a registration form

Here, we will create a form and store the data collected from it in our school database.

We will Create two files, **index.php** and **connect.php.**

The file **index.php** will carry the **form HTML code**, while **connect.php** connects the **form with the database.**

Save as index.php

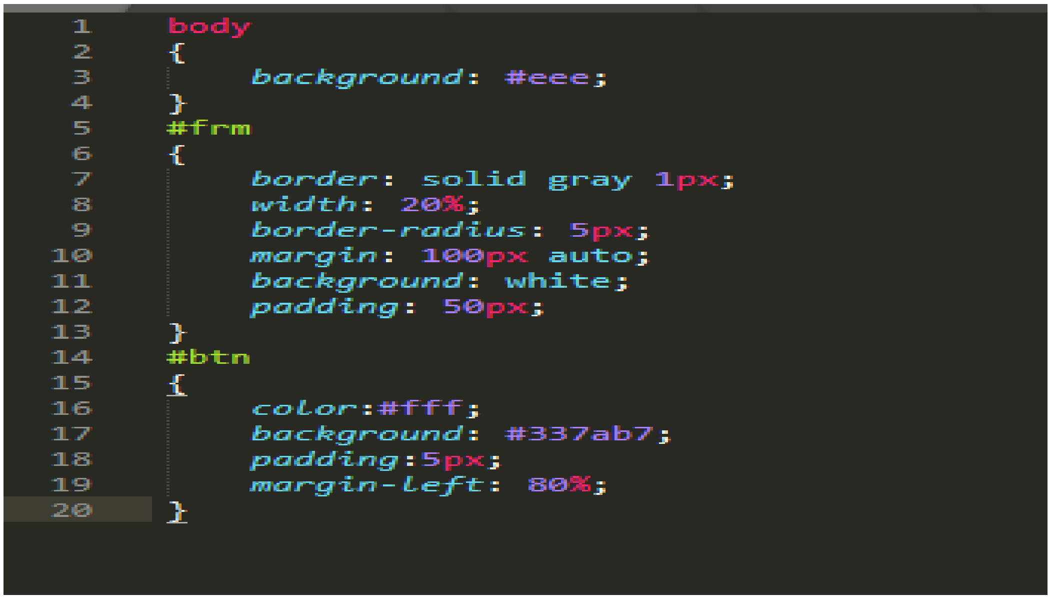


```
<!DOCTYPE html>
<html>
<head>
  <title>Working with the database </title>
<link rel="stylesheet"type="text/css"href="style.css">
</head>
<body>
<div id="frm">
  <h1>Registration Form</h1>
  <h3>Saving Data into the database</h3>
  <form action="connect.php" method="POST">
    <label>Firstname</label><br>
    <input type="text" placeholder="Enter firstname" name="firstname" required ="true">
    <br>
    <label>Lastname</label><br>
    <input type="text" placeholder="Enter lastname" name="lastname" required ="true">
    <br>
    <label>Email</label><br>
    <input type="email" placeholder="Enter email" name="email" required ="true">
    <br>
    <label>Phone</label><br>
    <input type="phone" placeholder="Enter Phone" name="phone" required ="true">
    <br>
    <input type="submit"id="btn"value="login"/>
  </form>
  </div>
</body>
</html>
```

Next document

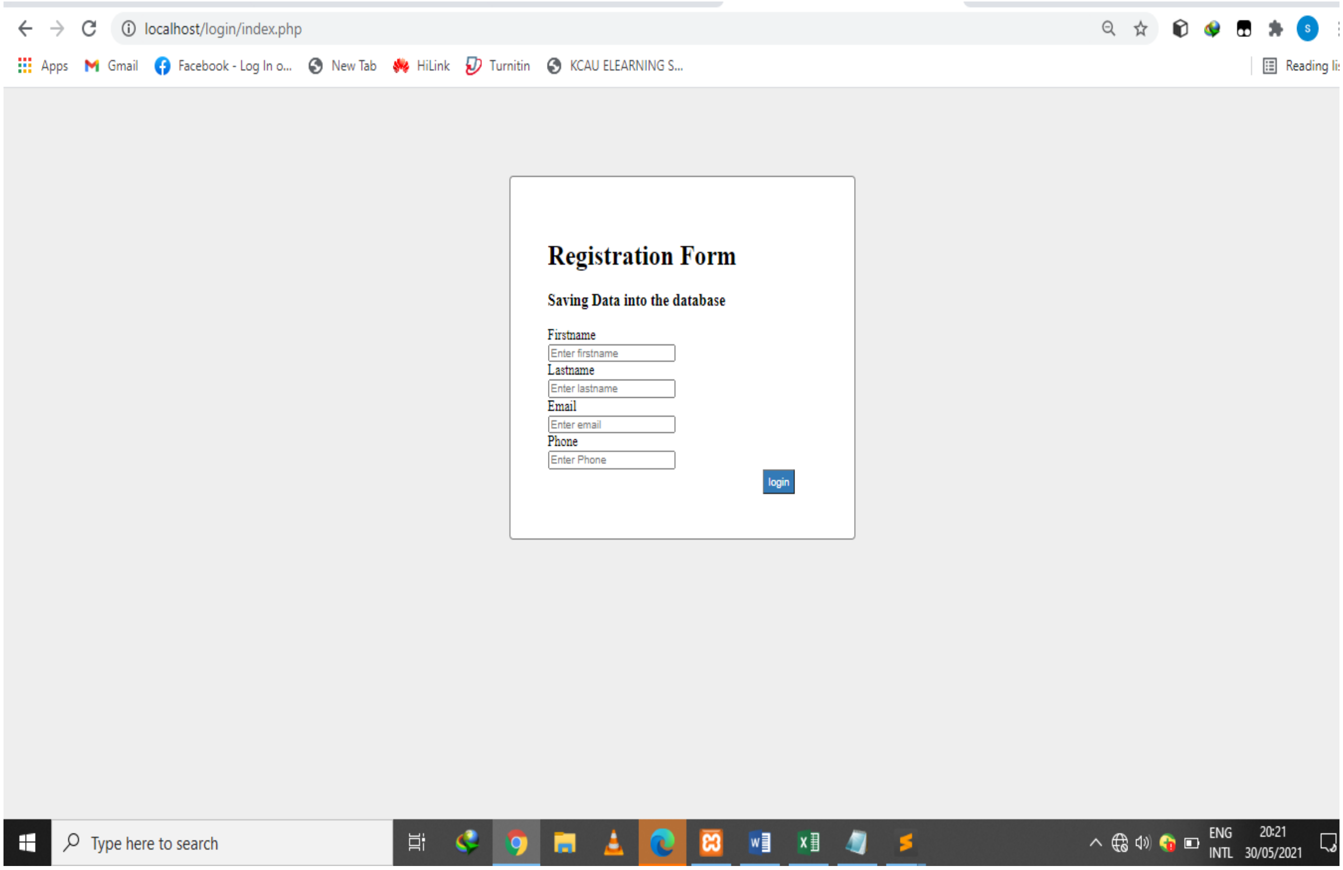
Style.css

Save as Style.css



```
body
{
	background: #eee;
}
#frm
{
	border: solid gray 1px;
	width: 20%;
	border-radius: 5px;
	margin: 100px auto;
	background: white;
	padding: 50px;
}
#btn
{
	color:#fff;
	background: #337ab7;
	padding:5px;
	margin-left: 80%;
}
```

The page should now look like this:


localhost/login/index.php
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Type here to search
ENG
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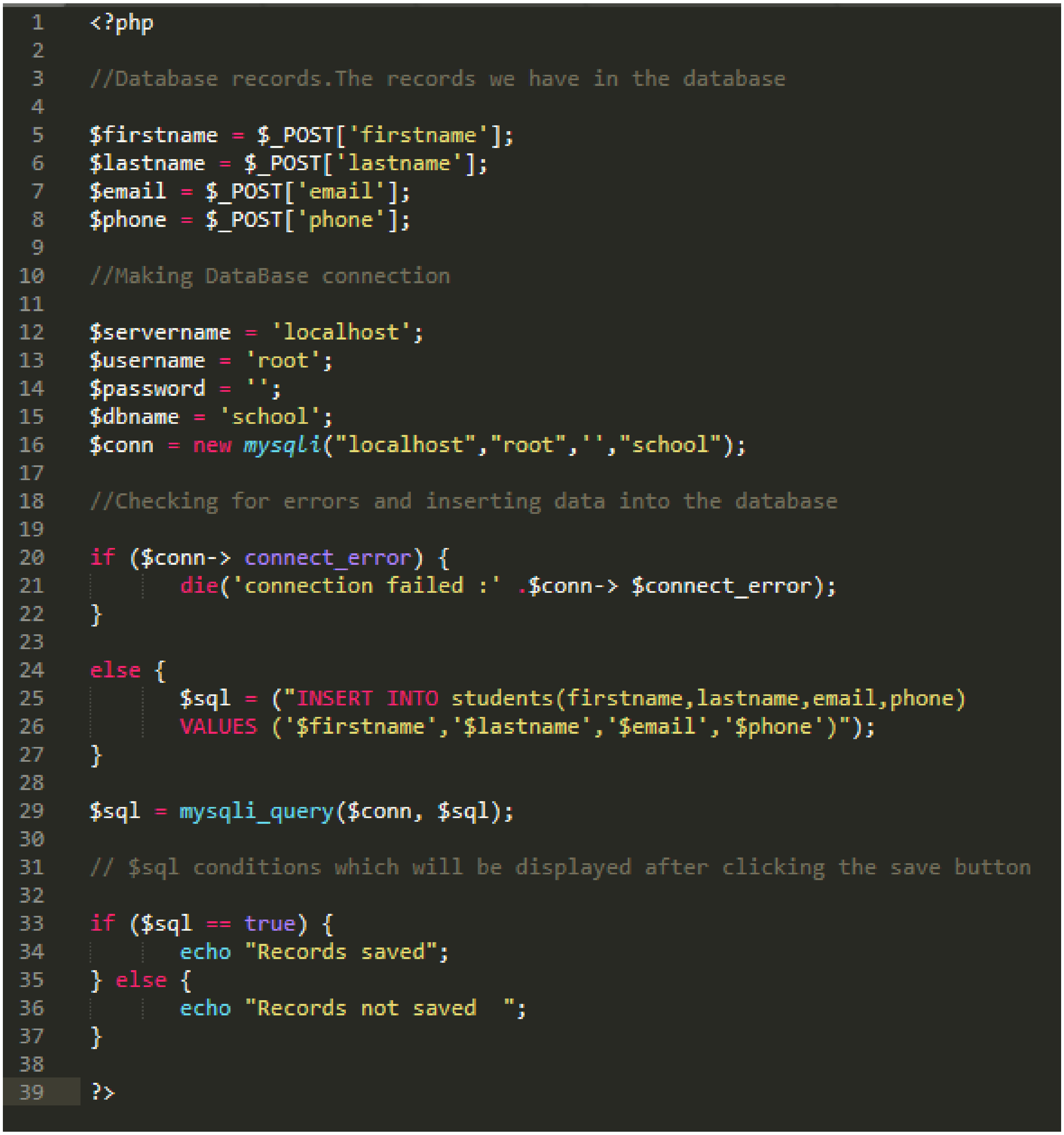
Registration Form

Saving Data into the database

Firstname
Enter firstname
Lastname
Enter lastname
Email
Enter email
Phone
Enter Phone
login

Now, let’s connect our form with the database.

Save as connect.php :

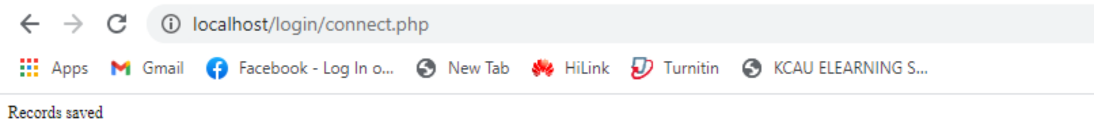


```php
<?php

//Database records.The records we have in the database

$firstname = $_POST['firstname'];
$lastname = $_POST['lastname'];
$email = $_POST['email'];
$phone = $_POST['phone'];

//Making DataBase connection

$servername = 'localhost';
$username = 'root';
$password = '';
$dbname = 'school';
$conn = new mysqli("localhost","root",'',"school");

//Checking for errors and inserting data into the database

if ($conn-> connect_error) {
		die('connection failed :' .$conn-> $connect_error);
}

else {
		$sql = ("INSERT INTO students(firstname,lastname,email,phone)
		VALUES ('$firstname','$lastname','$email','$phone')");
}

$sql = mysqli_query($conn, $sql);

// $sql conditions which will be displayed after clicking the save button

if ($sql == true) {
		echo "Records saved";
} else {
		echo "Records not saved  ";
}

?>
```

8.3 Storing data in the database

When we open http://localhost/phpmyadmin/ on the browser we should have something like this in our database.

- **Open index.php >fill in the registration form**
- **After filling the form >**click login **then if results are** **successful Record saved**


localhost/login/connect.php
Apps
Gmail
Facebook - Log In o...
New Tab
HiLink
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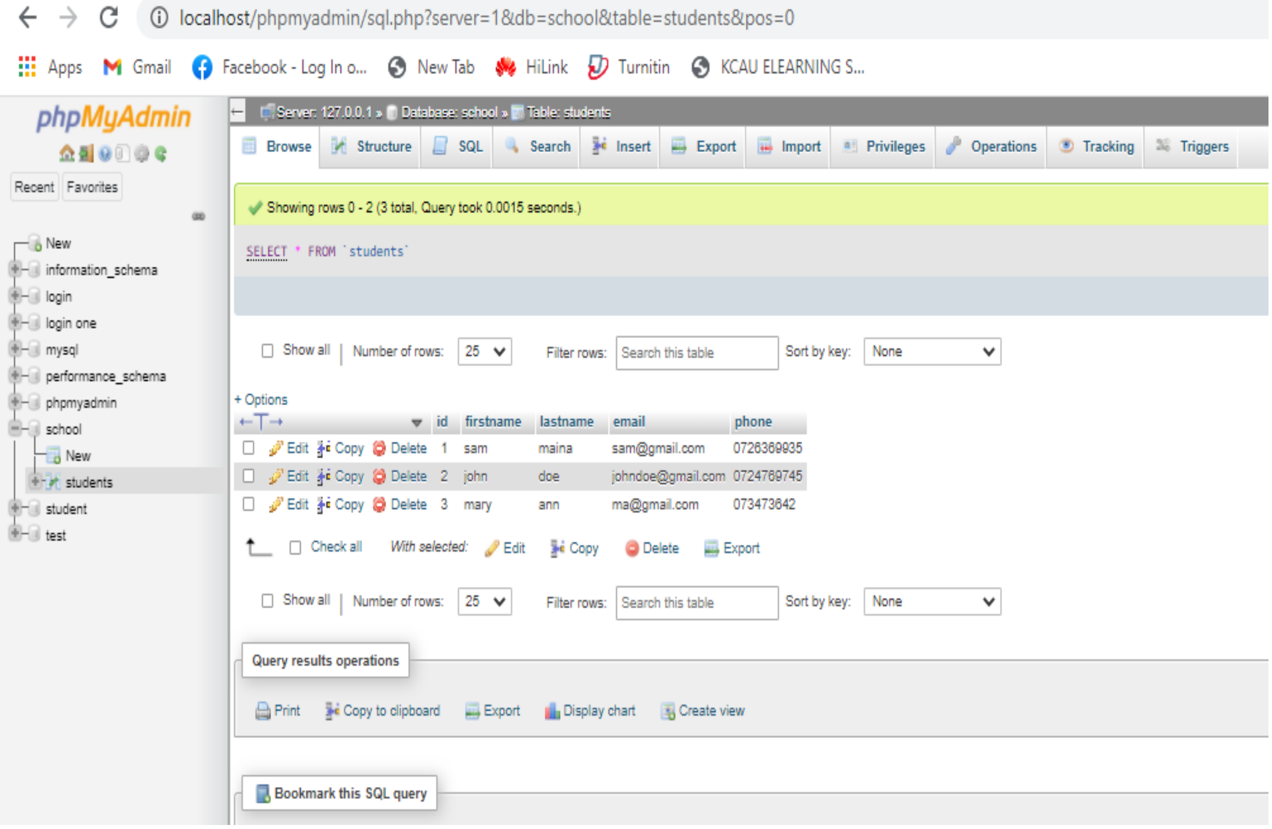
Records saved

- When we open http://localhost/phpmyadmin/ on the browser we should have something like this in our database.


localhost/phpmyadmin/sql.php?server=1&db=school&table=students&pos=0
Apps
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performance_schema
phpmyadmin
school
New
students
student
test
Server: 127.0.0.1 » Database: school » Table: students
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Tracking
Triggers
☐ Show all
Number of rows: 25
Filter rows: Search this table
Sort by key: None
+ Options
☐ Check all
With selected: Edit Copy Delete Export
Query results operations
Print
Copy to clipboard
Export
Display chart
Create view
Bookmark this SQL query

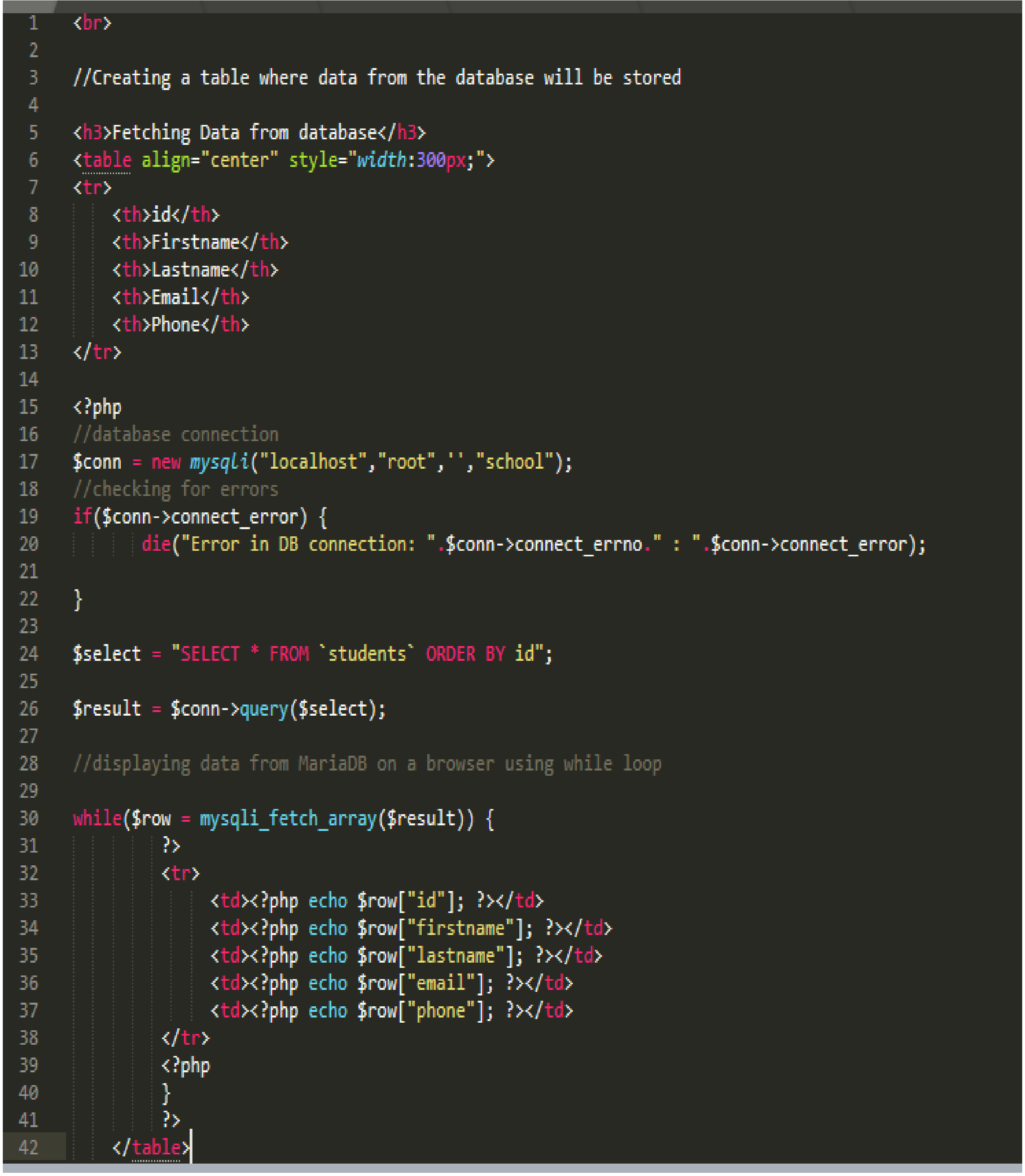
Showing rows 0 - 2 (3 total, Query took 0.0015 seconds.)

```
SELECT * FROM `students`
```

	id	firstname	lastname	email	phone
☐ Edit Copy Delete	1	sam	maina	sam@gmail.com	0726369935
☐ Edit Copy Delete	2	john	doe	johndoe@gmail.com	0724769745
☐ Edit Copy Delete	3	mary	ann	ma@gmail.com	073473642

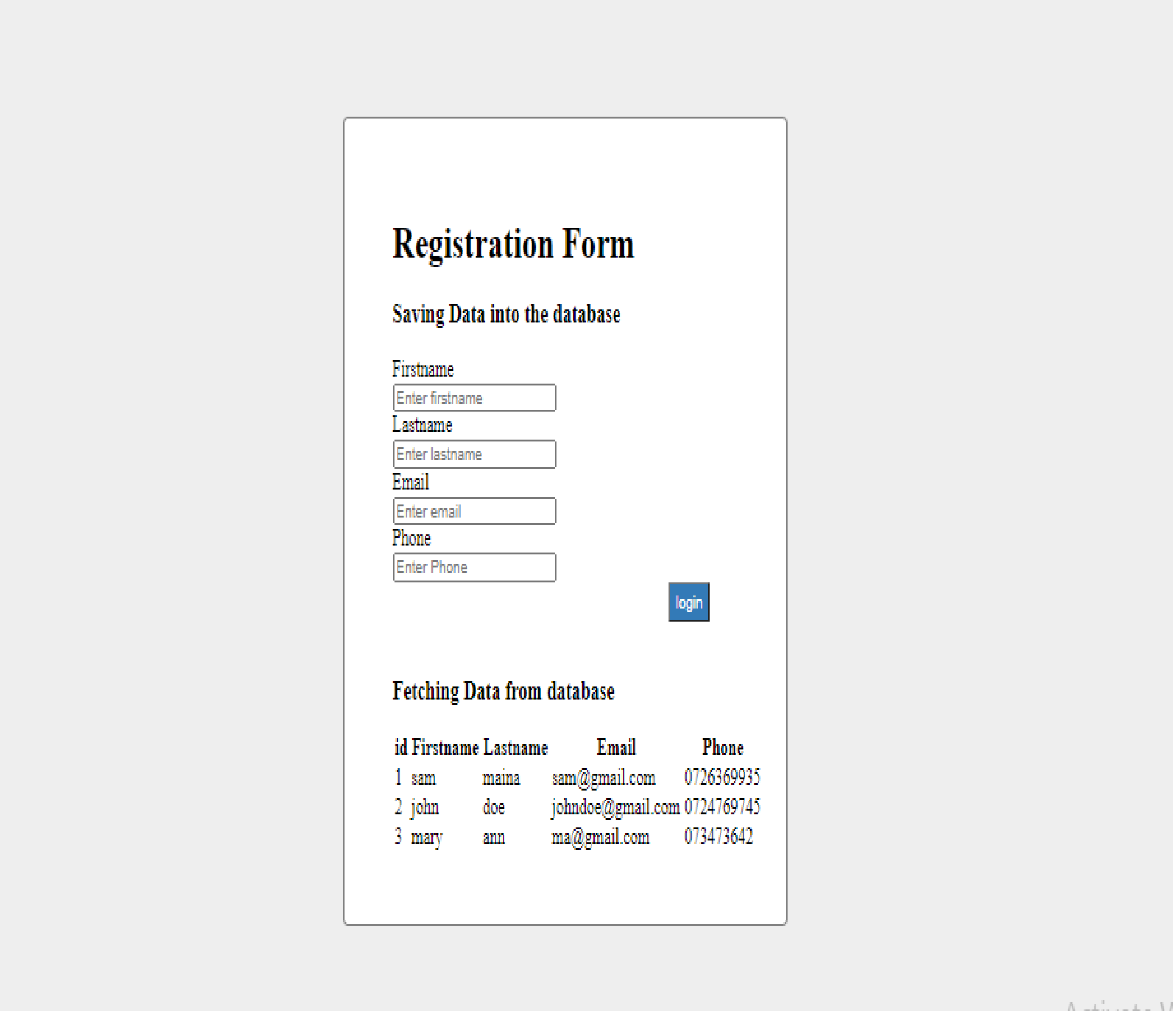
- ## 8.4 Fetching data from the Database (using CRUD operation) `Create, Read, Update &delete`

To fetch the data that we just saved, add the code below into `index.php` right under the `</form>` tag.



```
<br>

//Creating a table where data from the database will be stored

<h3>Fetching Data from database</h3>
<table align="center" style="width:300px;">
<tr>
    <th>id</th>
    <th>Firstname</th>
    <th>Lastname</th>
    <th>Email</th>
    <th>Phone</th>
</tr>

<?php
//database connection
$conn = new mysqli("localhost","root",'',"school");
//checking for errors
if($conn->connect_error) {
       die("Error in DB connection: ".$conn->connect_errno." : ".$conn->connect_error);

}

$select = "SELECT * FROM `students` ORDER BY id";

$result = $conn->query($select);

//displaying data from MariaDB on a browser using while loop

while($row = mysqli_fetch_array($result)) {
         ?>
         <tr>
              <td><?php echo $row["id"]; ?></td>
              <td><?php echo $row["firstname"]; ?></td>
              <td><?php echo $row["lastname"]; ?></td>
              <td><?php echo $row["email"]; ?></td>
              <td><?php echo $row["phone"]; ?></td>
         </tr>
         <?php
         }
         ?>
    </table>
```

On refreshing the page, we should see:



Registration Form

Saving Data into the database

Firstname
Enter firstname
Lastname
Enter lastname
Email
Enter email
Phone
Enter Phone
login

Fetching Data from database

id	Firstname	Lastname	Email	Phone
1	sam	maina	sam@gmail.com	0726369935
2	john	doe	johndoe@gmail.com	0724769745
3	mary	ann	ma@gmail.com	073473642

The data at the bottom of the page has been fetched from the database.